Lynbrook Math Summer Camp 2026
N2 Divisibility

Lynbrook Math Club

August 2026

N2.1 Lecture

When working with integers, checking whether one number divides another is an important skill. While long division always works, it is often way too slow when dealing with large numbers or missing digits.

In this lesson, we will explore faster divisibility tests and clever algebraic tricks that allow us to determine divisibility instantly.

Fundamental Properties of Divisibility

Note that for integers a and b (with $a \neq 0$), we write $a \mid b$ to mean that a divides b. Before testing specific numbers, we must establish a few essential rules governing how divisibility interacts with addition and multiplication.

> **Claim N2.1.1** (Divisibility Linear Combination) — If $a \mid b$ and $a \mid c$, then $a \mid (xb + yc)$ for any integers x and y.

This property is extremely powerful: if a number divides two separate quantities, it must also divide any sum or difference formed by those quantities.

Last Digit Divisibility Rules

Our base-10 number system gives us some easy shortcut tests for powers of 2 and 5. Because $10 = 2 \times 5$, higher powers of 10 are divisible by powers of 2 and 5.

Definition N2.1.2 (Last Digit Tests)**.** Let N be a positive integer written in base 10.

- N is divisible by 2 or 5 if and only if its **last** 1 **digit** is divisible by 2 or 5, respectively.
- N is divisible by 4 or 25 if and only if its **last** 2 **digits** form a number divisible by 4 or 25.
- N is divisible by 8 or 125 if and only if its **last** 3 **digits** form a number divisible by 8 or 125.

Why does this work? Notice that any number N can be split into a multiple of 100 plus its last two digits:

$$N = 100k + \overline{ab}$$

Since $4 \mid 100k$ for every integer k, N is divisible by 4 if and only if $4 \mid \overline{ab}$. The same logic applies to 8 because $8 \mid 1000k$.

Example 3

Find the single digit d such that the five-digit number $5432d$ is divisible by 8.

By the 8 divisibility rule, we only need to look at the last three digits, $\overline{32d}$, which represents the number $320 + d$.

Since $320 = 8 \times 40$, 320 is already divisible by 8. Therefore, $320 + d$ is divisible by 8 if and only if d is divisible by 8. The only single-digit values possible are $d = 0$ or $d = 8$.

Digit Sum Rules (3 and 9)

The divisibility tests for 3 and 9 depend on the sum of a number's digits rather than its positional end.

Claim N2.1.4 (Sum of Digits Rule) — A positive integer N is divisible by 3 (or 9) if and only if the **sum of its digits** is divisible by 3 (or 9).

To see why, notice that $10 = 9 + 1$, $100 = 99 + 1$, and in general, $10^k = (\text{a multiple of } 9) + 1$. If N has digits $a_n a_{n-1} \dots a_1 a_0$, we can express N as:

$$N = a_n(10^n) + a_{n-1}(10^{n-1}) + \cdots + a_1(10) + a_0$$

Replacing each power 10^k with $(\text{multiple of } 9 + 1)$, N becomes:

$$N = (\text{multiple of } 9) + (a_n + a_{n-1} + \cdots + a_1 + a_0)$$

Thus, N leaves the exact same remainder as its digit sum when divided by 3 or 9.

Example 5

The 7-digit number $32A514B$ is a multiple of 9. If $A - B = 3$, find the pair (A, B).

First, calculate the sum of the digits:

$$S = 3 + 2 + A + 5 + 1 + 4 + B = 15 + A + B$$

For the number to be a multiple of 9, $15 + A + B$ must be a multiple of 9. Since A and B are single digits ($0 \leq A, B \leq 9$), $A + B$ can range from 0 to 18.

The possible multiples of 9 for $15 + A + B$ are 18 and 27:

- If $15 + A + B = 18 \implies A + B = 3$.
- If $15 + A + B = 27 \implies A + B = 12$.

We are given $A - B = 3$.

- **Case 1:** $A + B = 3$ and $A - B = 3$. Adding the equations gives $2A = 6 \implies A = 3$, which leaves $B = 0$.
- **Case 2:** $A + B = 12$ and $A - B = 3$. Adding gives $2A = 15 \implies A = 7.5$, which is impossible since A must be an integer.

Thus, the unique solution is $(A, B) = (3, 0)$.

Alternating Digit Sum (11)

Claim N2.1.6 (Divisibility by 11) — A number is divisible by 11 if and only if its **alternating sum of digits** (subtracting and adding digits alternately from right to left) is divisible by 11.

This works because powers of 10 alternate in relation to 11:

$$10 = 11 - 1, \quad 100 = 99 + 1, \quad 1000 = 1001 - 1, \quad \dots$$

In general, $10^k = (\text{multiple of } 11) + (-1)^k$.

Example 7

What is the missing digit x if $7x4321$ is divisible by 11?

We calculate the alternating sum starting from the rightmost digit:

$$1 - 2 + 3 - 4 + x - 7 = x - 9$$

For $x - 9$ to be divisible by 11, $x - 9$ must be a multiple of 11. Since $0 \leq x \leq 9$, the only valid choice is $x - 9 = 0 \implies x = 9$.

Composite Divisors

To test divisibility by a composite number, we decompose it into prime factors.

Theorem N2.1.8 (Coprime Split Rule)

If a and b are relatively prime (meaning $\gcd(a, b) = 1$), then $ab \mid N$ if and only if $a \mid N$ and $b \mid N$.

Remark N2.1.9. This rule **only** works if a and b share no common factors! For example, to check if a number is divisible by 6, we check if it is divisible by 2 and 3. However, to check if a number is divisible by 12, we test 3 and 4 (not 2 and 6, since $\gcd(2, 6) \neq 1$).

Example 10

Find the number of pairs of digits (x, y) such that the six-digit number $4x51y2$ is divisible by 72.

Since $72 = 8 \times 9$ and $\gcd(8, 9) = 1$, $4x51y2$ must be divisible by both 8 and 9.

1. **Test for 8:** The last three digits $\overline{1y2} = 100 + 10y + 2 = 102 + 10y$ must be divisible by 8. Testing possible values of $y \in \{0, 1, \dots, 9\}$:

- $y = 1 \implies 112 = 8 \times 14$ (Works)
- $y = 5 \implies 152 = 8 \times 19$ (Works)
- $y = 9 \implies 192 = 8 \times 24$ (Works)

So y can be 1, 5, or 9.

2. **Test for 9:** The digit sum $4 + x + 5 + 1 + y + 2 = 12 + x + y$ must be a multiple of 9.

- **If** $y = 1$**:** $12 + x + 1 = 13 + x \implies x = 5$.
- **If** $y = 5$**:** $12 + x + 5 = 17 + x \implies x = 1$.

- **If** $y = 9$**:** $12 + x + 9 = 21 + x \implies x = 6$.

Thus, there are 3 valid pairs: $(5, 1)$, $(1, 5)$, and $(6, 9)$.

Algebraic Tricks for Divisibility

Sometimes, divisibility questions involve expressions rather than explicit digits. Two algebraic identities are especially handy for these problems.

Claim N2.1.11 (Useful Factorization Identities) — For any positive integer n:

- $a - b \mid a^n - b^n$ for all n.
- $a + b \mid a^n + b^n$ for all **odd** n.

Example 12
Show that $2^{3n} - 1$ is always divisible by 7 for any positive integer n.

Using exponent rules, we can rewrite the expression as:

$$2^{3n} - 1 = (2^3)^n - 1^n = 8^n - 1^n$$

Applying our identity with $a = 8$ and $b = 1$, we know that $(a - b) \mid (a^n - b^n)$:

$$(8 - 1) \mid (8^n - 1^n) \implies 7 \mid (2^{3n} - 1)$$

This completes the proof.